Immunology

Ingyu Bahng

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Contents

1 Immune System Players

Core Concept 1.1: Hematopoietic Stem Cells (HSCs) & Lineage

Hematopoietic stem cells (HSCs) are pluripotent stem cells capable of giving rise to any type of blood cell. Each HSC can either divide to produce additional copies of itself—a process termed self-renewal—or begin to differentiate through a series of lineage-restricting choices.

Developmentally, HSCs first arise in the yolk sac membrane of the early embryo, subsequently migrate to the liver and spleen, and ultimately settle predominantly in the **bone marrow** prior to birth. Remarkably, as few as approximately 100 HSCs are sufficient to fully regenerate the entire hematopoietic system.

Figure 1 illustrates the hierarchical differentiation of HSCs into the myeloid and lymphoid lineages, highlighting the key transcription factors governing lineage commitment at each branch point.

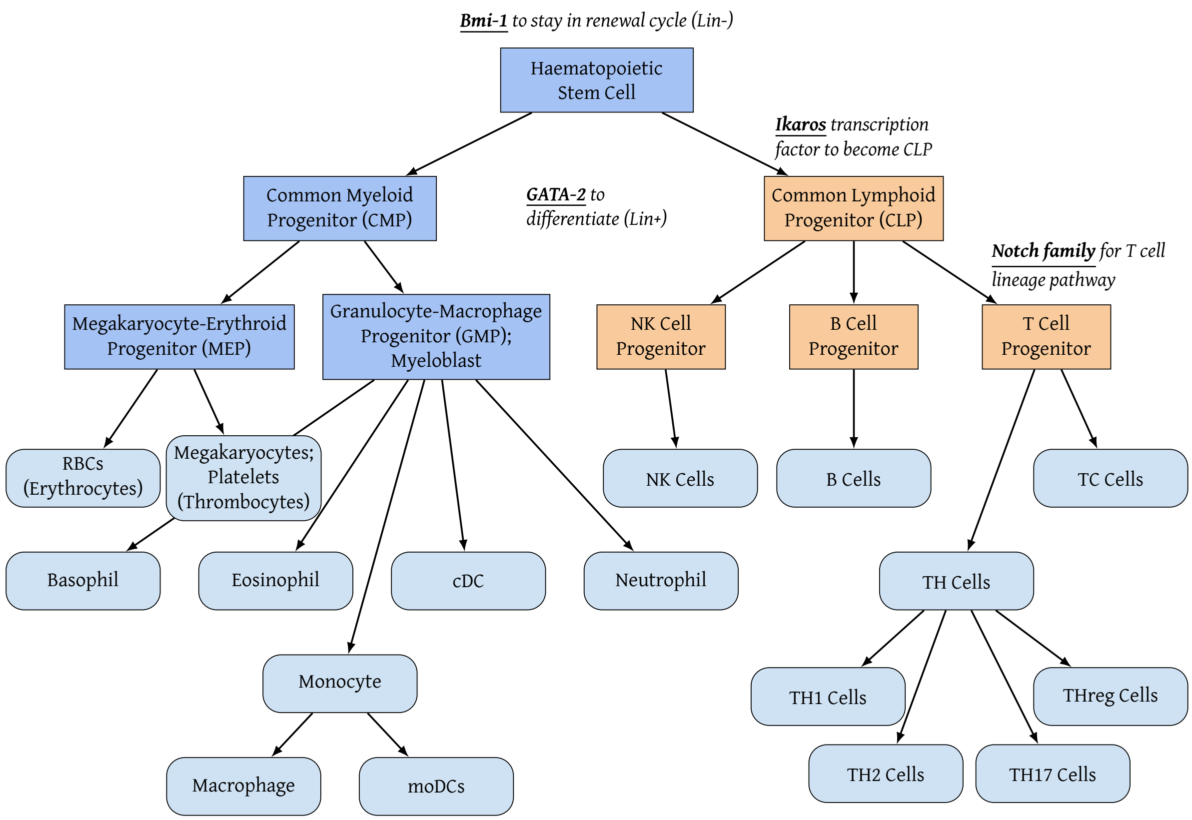

Figure 1: Hierarchy of haematopoietic stem cell differentiation into myeloid (*left*) and lymphoid (*right*) lineages

As shown in Figure 1, HSCs destined for differentiation are termed **lineage positive (*lin+*)**, whereas those not destined for differentiation are termed **lineage negative (*lin-*)**. This binary state is governed by the presence of one of two transcription factors.

- **Bmi-1**: A transcription factor whose presence maintains HSCs in an **undifferentiated state**, sustaining ongoing self-renewal and division.
- **GATA-2**: A transcription factor that **triggers HSC differentiation**, initiating the developmental path toward a specialized cell type.

Given a *lin+* state, the first branch point determines whether the cell commits to the myeloid or lymphoid lineage.

- Commitment to a **common myeloid-erythroid progenitor** enables development into a wide array of cell types, including various white blood cells, red blood cells, and platelet-producing megakaryocytes.
- Commitment to a **common lymphoid progenitor** is marked by expression of the transcription factor **Ikaros**, giving rise to adaptive T and B cells or innate NK and dendritic cells. Further along the differentiation tree, the **Notch family** of transcription factors governs the divergence of T and B cell development.

Core Concept 1.2: Lymphoid Lineage

Cells of the **lymphoid lineage** constitute 20–40% of circulating leukocytes in the blood and approximately 99% of the cells of the lymph. These cells are broadly categorized as either **adaptive**—characterized by gene rearrangement during the immune response—or **innate**—characterized by the absence of such gene rearrangement.

Table 1 presents a structured overview of the cell types comprising the lymphoid lineage depicted in Figure 1.

Core Concept 1.3: Myeloid Lineage

Cells of the **myeloid lineage** can be broadly categorized into **granulocytic cells**—characterized by specific granules that compartmentalize potentially dangerous molecules—and **antigen-presenting cells (APCs)**, which present antigen to T cells. Additional myeloid lineage cell types include **red blood cells (*erythrocytes*)** and **platelets (*thrombocytes*)**, which serve more secondary roles within general immune function.

Table 2 presents a structured overview of the cell types comprising the myeloid lineage depicted in Figure 1.

Core Concept 1.4: Follicular Dendritic Cells

Follicular dendritic cells (FDCs) are specialized stromal cells that, in contrast to the myeloid, antigen presenting **sentinel dendritic cells** shown in Table 2, do not arise via **hematopoiesis** from HSCs. Instead, they originate during embryogenesis from a distinct **mesenchymal** lineage of **mesenchymal progenitor cells**.

Following development, FDCs migrate into lymphoid tissues, where they ultimately become embedded within the follicles of secondary lymphoid organs.

Once established, FDCs are distinguished by their unique capacity to **retain and present intact antigens** on their surfaces over extended periods. They capture antigens from the lymph and retain them as immune complexes, a function essential for **B cell activation** and **germinal center formation**.

The most characteristic surface markers of FDCs are as follows.

- **CD21/CD35**: FDCs express the complement receptors CD21 and CD35, which mediate their role in antigen capture.
- **FDC-M1**: This marker is specific to FDCs, enabling their identification and distinction from other cellular populations within lymphoid tissue.

Table 1: Overview of adaptive and innate lymphocyte populations

Type	Name	Function	Surface Expression	Responds to	Releases	Notes
ADAPTIVE	*T_H (Helper) Lymphocyte*	**Activates** inflammation, macrophages, and B cells	[CD4]—stabilizes MHC II–TCR antigen recognition [TCR]	MHC II-presented antigen	Cytokines	Subtypes include: **T_H1**—direct response **T_H2**—nutrient restraint response **T_H17**—fungi/parasite response **Memory cells**—future reserve
	T_R (Regulatory) Lymphocyte	**Suppresses** immune response (*generally and of specific lymphocytes*)	[CD4] [CD25]—stabilizes T_{reg} binding to IL-2, essential for survival and function	*Unknown*	Cytokines; IL-10; TGF-β	Prevents allergy and autoimmune disease; subtype of the TH lineage
	T_C (Cytotoxic) Lymphocyte	**Destroys** altered-self cells	[CD8]—stabilizes MHC I–TCR antigen recognition	MHC I-presented antigen	Perforin; Granzymes	
	B Cell	**Recognizes** antigen; can present antigen to T cells	[Membrane-bound antibodies/BCRs] [MHC II]—constitutive; presents BCR-captured antigen fragments [TLR] [Fc receptors]—down-regulate antibody production [B7]	Circulating antigen; T_H cytokine signals	[B7]	Develops into a **plasma cell** or **memory cell** (*reserve for future infection*)
	Plasma Cell	**Secretes** antibodies	No surface antibodies	T_H cytokine signals	Antibodies	Mature B cell; Lives 1–2 weeks
INNATE	*NK (Natural Killer) Cell*	**Destroys** altered-self cells	[CD16]—Fc receptor recognizing antibody-coated altered cells, triggering attack [MHC I receptors]—induce killing when MHC I is absent	Antibodies; downregulated MHC I	Perforin; Granzymes	Executes **antibody-dependent cell-mediated cytotoxicity (ADCC)**, triggering apoptosis via antibody signals on antigen-coated cell surfaces

Table 2: Overview of innate granulocytic (*1st page*), innate antigen presenting (*2nd page*), and other myeloid cell populations (*2nd page*)

Type	Name	Function	Surface Expression	Responds to	Releases	Notes
INNATE (Granulocytic)	*Neutrophil*	**Phagocytosis**, mediates the acute inflammatory response Migrates into infected tissues	[Fc Receptors]—binds to the **fragment crystallizable** region of antibodies, facilitating the recognition of antibody-coated pathogens and subsequent immune activation [Complement Receptors]—enhances the phagocytosis of pathogens opsonized by complement proteins [PRR]—mediates pathogen pattern recognition; specifically **Toll-like receptors**	Chemokines	Acidic & basic granules ROS/RNS via the NOX complex [IL-1] [IL-6] [TNFα]	First responders to inflection One-day lifespan Multilobed nucleus
	Basophil	**Defense** against allergic responses and parasitic infections	[Fc Receptors]—for IgE antibodies	*Chemokines*	Histamine-rich basophilic granules	Non-phagocytic Lobed nucleus Responds to worms
	Mast Cell	**Defense** against allergic responses and parasitic infections	[Fc Receptors]—for IgE antibodies		Histamine-rich basophilic granules	Non-phagocytic Mononuclear (non-lobed)
	Eosinophil	**Defense** against allergic responses and parasitic infections	[Fc Receptors]—for IgE antibodies	*Chemokines*	Eosinophilic granules containing hydrolytic enzymes	Phagocytic, though not its primary function Bi-lobed nucleus

continued on next page

Type	Name	Function	Surface Expression	Responds to	Releases	Notes
INNATE (Antigen Presenting)	*Sentinel Dendritic Cell (DC)*	**Processes and presents** antigens to T_H cells *Minor roles:* **Phagocytosis**, **endocytosis**, and **pinocytosis** of microorganisms and cellular debris	[MHC II]—constitutive [B7]—facilitates T cell activation [PRR]—mediates pathogen pattern recognition; specifically **Toll-like receptors**		[B7]—co-stimulatory molecule provided to T cells during MHC-TCR antigen presentation	DCs can differentiate from CMPs, monocytes, and CLPs (*each yielding distinct specialized subsets*) Reside in peripheral tissues, **migrating to secondary lymphoid organs** only upon detecting pathogenic signals
	Monocyte	**Precursors** to macrophages, DCs, and MDSCs	[TLR]	Interferon from T_H cells		Circulates in the bloodstream for approximately 8 hours
	Macrophage	**Phagocytosis** of microorganisms and cellular debris *Minor role:* **Presents** antigens following activation	[MHC II]—post-activation [PRR]—mediates pathogen pattern recognition; specifically **Toll-like receptors**		ROS/RNS via the NOX complex [B7] [IL-1],[IL-6] [TNFα]	Exhibits amoeboid motility Pathogens opsonized by antibodies are more readily phagocytized
OTHER	*Red Blood Cells (RBC; Erythrocytes)*	**Distributes** oxygen and nutrients systemically	[CD235a]—contributes to RBC antigenic properties			
	Platelets (Thrombocytes)	**Mediates** coagulation at sites of vascular injury	[CD41]—functions as a key marker essential for platelet aggregation [CD42b]—facilitates platelet adhesion via a glycoprotein complex			Platelets are derived from megakaryocytes

2 Immune System Organs

2.1 Primary Immune System Organs

Core Concept 2.1: Primary Immune System Organs

The **primary immune system organs** are the sites at which cells divide, commit to a developmental fate, and, where applicable, undergo gene rearrangement.

Core Concept 2.2: Bone Marrow

The **bone marrow** is the soft, highly vascularized tissue residing within the hollow interior cavities of bones. It is the primary site of **hematopoietic stem cells (HSCs)** (*Core Concept 1.1*) and continuous blood cell generation in adult mammals, serving as the location for the complete maturation of all myeloid cells, NK cells, and B cells, as well as the site of the initial developmental stages of T cells.

The major anatomical sites containing active bone marrow include the femur, humerus, hip bones, and sternum.

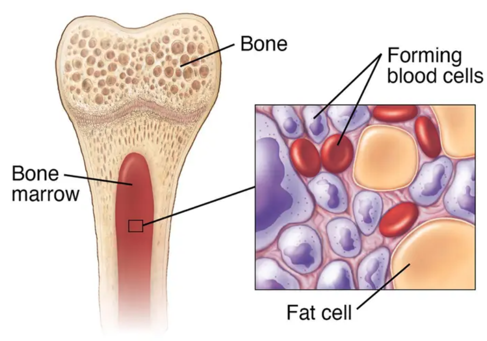

Figure 2: Bone marrow

Core Concept 2.3: Thymus

The **thymus**, a small gland located above the heart, is the primary site of **T cell maturation** and development, involving a rigorous, stepwise selection process that tests for the correct functionality of T cells and their surface receptors. Cells that successfully survive this selection process enter systemic circulation, whereas the vast majority that fail undergo **apoptosis**.

Structurally, the thymus is organized into an **outer cortex** and an **inner medulla**, each supported by a distinct network of epithelial cells that guide developing **thymocytes** (*immature, developing T cells*) through selection.

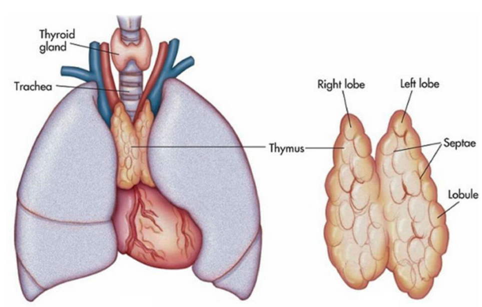

Figure 3: Thymus

2.2 Secondary Immune System Organs

Core Concept 2.4: Secondary Immune System Organs

The **secondary immune system organs** are the sites at which information regarding pathogens is coordinated, leading to the subsequent activation of immune cells.

Core Concept 2.5: Lymphatic System (Fluids, Vessels, Nodes)

The **lymphatic system** functions as a **unidirectional** transport network, carrying fluid from peripheral tissues back toward the heart and ultimately the bloodstream.

- **Vessel & Fluid**

  Lymphatic vessels drain interstitial fluid, collecting antigens and leukocytes in the process; they facilitate the transport of immune cells but, unlike blood vessels, are devoid of **red blood cells (RBCs)**. Lymph fluid undergoes filtration as it passes through the lymph nodes.

- **Lymph Nodes**

  Lymph nodes serve primarily to [1] **trap incoming antigens** presented by APCs and [2] **provide a localized environment for lymphocytes**—T and B cells—to interact with these antigens, thereby generating a subsequent adaptive immune response.

  The resulting activated lymphocytes exit the node through the efferent lymphatic vessels, ultimately entering the bloodstream via the thoracic duct and subclavian vein to traffic toward the site of infection.

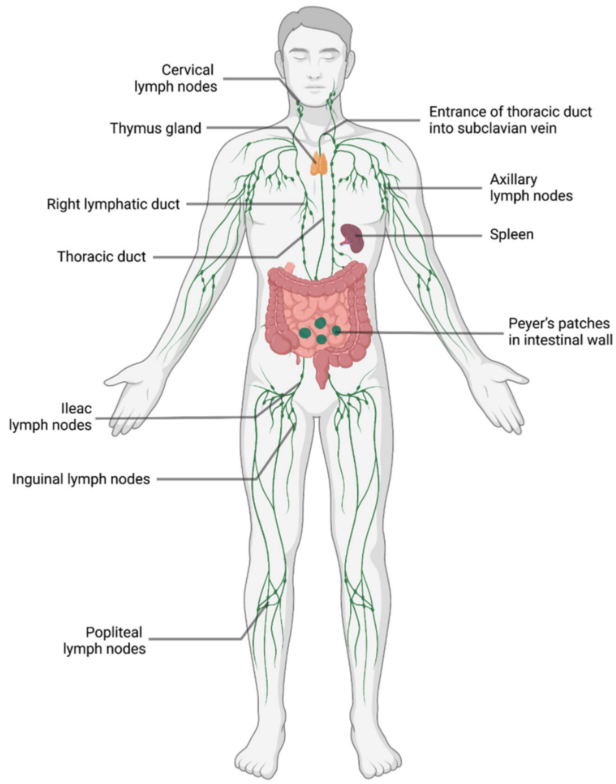

Figure 4: Lymphatic system

Core Concept 2.6: Spleen

The **spleen** functions strictly to **filter blood**, as opposed to lymph, and can be categorized into two tissue types.

- **Red Pulp** is responsible for recycling senescent RBCs and platelets from circulation via macrophage-mediated phagocytosis.
- **White Pulp** is a crucial site for immune activation, organized into **periarteriolar lymphoid sheaths (PALS)** primarily containing T cells, and marginal **lymphoid follicles** primarily containing B cells.

Surgical removal of the spleen can significantly elevate an individual's susceptibility to bacterial infections.

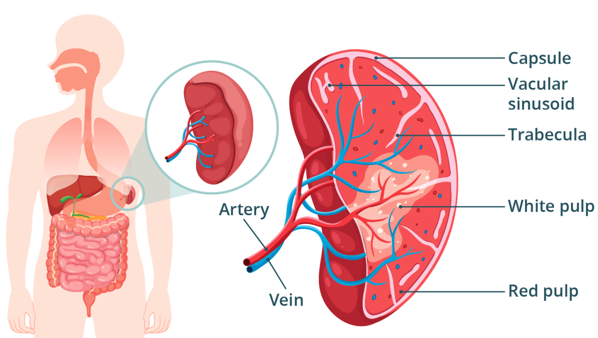

Figure 5: Spleen

Core Concept 2.7: Mucosa Associated Lymphoid Tissues (MALT)

Mucosa associated lymphoid tissues (MALT) refer to the collection of lymphoid tissues situated across various mucosal surfaces, including the gastrointestinal (GALT), bronchial (BALT), and nasal (NALT) tracts; these mucosal systems represent the primary sites of pathogen entry into the body.

Highly organized MALT structures include the tonsils, the appendix, and ***Peyer's patches*** *located in the intestinal tract.*

Microfold (M) cells (*Figure 6*) are specialized, large epithelial cells located in areas of localized lymphoid concentration throughout the MALT, each possessing a basolateral pocket that houses several smaller immune cells.

The epithelial cells of the mucosa serve the following functions.

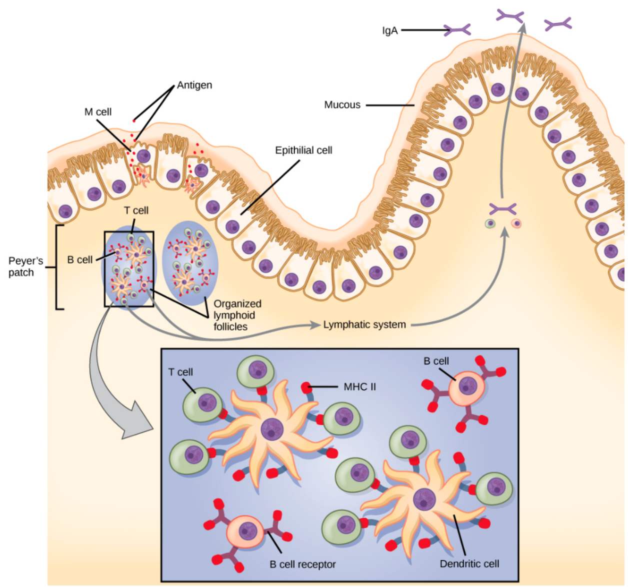

Figure 6: Mucosa associated lymphoid tissues; *microfold cell present towards the left end of epithelial cell line*

- They facilitate the **transport of antigen samples** from the **lumen** (*the space within vessels*) across the epithelial barrier via the specialized M cells.
- They **secrete fluids** such as tears and mucus, which contain antimicrobial enzymes like lysozyme that degrade bacterial membranes.
- They utilize ciliary action to **physically sweep away pathogens and debris**, a mechanism prominent in the respiratory and reproductive tracts.

2.3 Tertiary Immune System Organs

Core Concept 2.8: Tertiary Immune System Organs

A **tertiary immune organ** is an organized, non-encapsulated aggregate of immune and stromal cells that develops postnatally within non-lymphoid tissues at sites of persistent antigen exposure.

Core Concept 2.9: Skin

The **skin** is the largest organ of the human body and provides both innate and adaptive immune defense mechanisms. Its innate defense characteristics are as follows.

- **Keratinocytes**, the epithelial cells forming the outer epidermal layer, secrete **cytokines** to activate localized innate immune cells upon detecting a threat.
- Upon cell death, keratinocytes leave behind a protective **physical barrier** composed of keratin intermediate filaments.
- In the **epidermis**, sebaceous glands and hair follicles secrete **sebum**, antibacterial peptides, and **psoriasin** (*a specific antimicrobial peptide*) to eliminate pathogenic bacteria while maintaining normal flora.

- The underlying **dermis** consists of a deeper, thicker layer of mesenchymal and connective tissue that provides essential **blood supply and structural support**.

Conversely, the adaptive defense mechanisms of the skin are as follows.

- Keratinocytes, which are capable of expressing MHC II receptors (*Core Concept 7.1*), and **Langerhans cells** (*specialized dendritic cells of the skin*) present antigens to activate T_H cells.
- Additionally, the skin harbors **intra-epidermal lymphocytes**, a specialized population of resident T cells that continuously patrol the tissue.

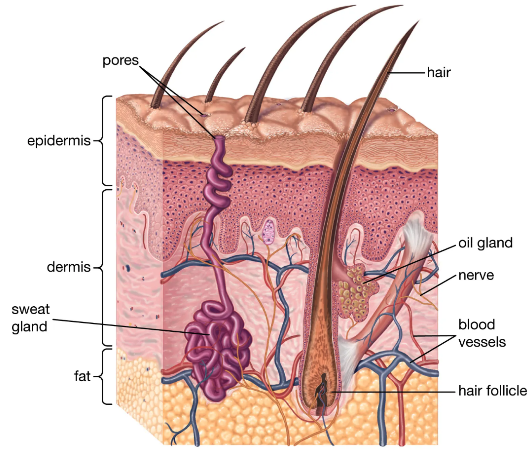

Figure 7: Skin

3 Innate Immunity Processes

Core Concept 3.1: Innate vs Adaptive Immune Response

The **innate immune response** is the body's first line of defense against pathogens. It is a rapid, **non-specific**, and evolutionary ancient defense mechanism that acts within minutes to hours of exposure. It is one that you can produce prior to exposure by a specific pathogen and doesn't require change to DNA/genes. All organisms have a form of innate immunity.

By contrast, the **adaptive response** is is the antigen-specific branch of the immune system that mounts a **tailored** defense against specific pathogen and creates long-lasting immunological memory. This response type takes days to develop during a primary infection because it requres the activation, gene rearrangement, clonal selection, and proliferation of lymphocytes with uniquely formed antigen receptors.

3.1 Inflammation

Core Concept 3.2: Inflammation

Inflammation represents a primary, coordinated physiological response that mobilizes systemic immune components to address and neutralize emerging pathogenic threats.

Core Concept 3.3: Fever & The Local Response

Fever is a key characteristic of the inflammatory response, induced by the **hypothalamus**, which regulates the body's baseline temperature. It serves as an overt systemic indicator of immune up-regulation, a response historically correlated with improved survival rates during infection.

Any form of tissue injury, including internal trauma, inherently triggers a **localized inflammatory response**, characterized by the classical signs of **swelling** (*due to vascular leakage*), **redness** (*due to increased local blood flow*), **localized heat**, and **pain**. Collectively, these physiological changes serve to discourage behaviors that might exacerbate the initial injury.

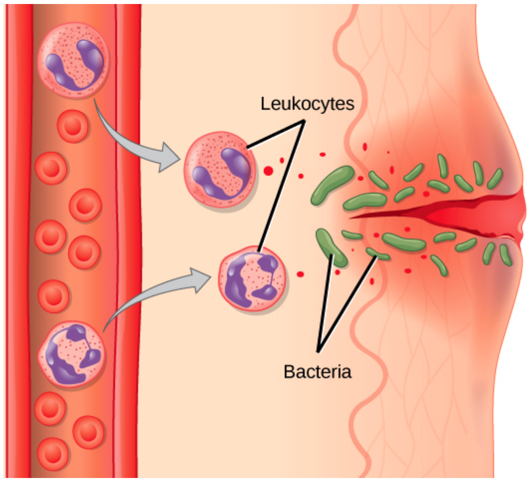

Figure 8: Localized inflammatory response; *invading pathogens and physical tissue damage from a skin laceration elicit the classical signs of inflammation and the recruitment of immune cells to defend against infection*

Core Concept 3.4: Immune Cell Extravasation

Extravasation is the process by which immune cells exit the circulatory system, migrating across the endothelial lining of blood vessels and into surrounding tissue toward a site of infection, injury, or inflammation. This process is particularly relevant to **neutrophils**, which are largely confined to the vasculature prior to infection and are selectively recruited to tissues via extravasation.

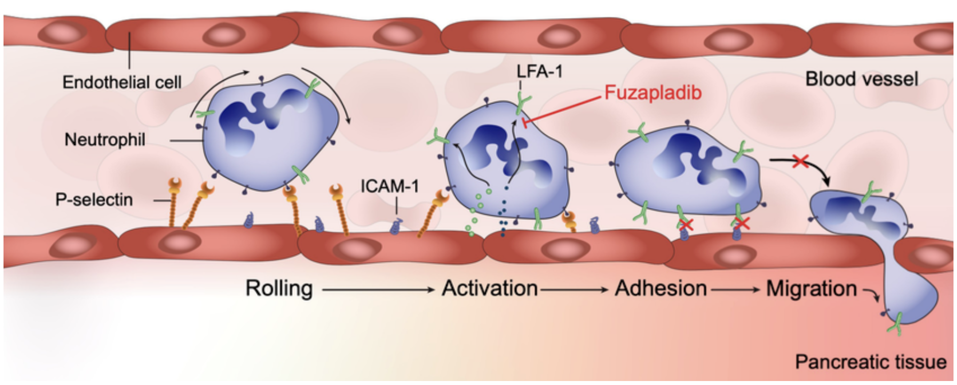

Figure 9: Extravastion of a neutrophil

(1) **Rolling**: Neutrophils form transient, low-affinity interactions with the endothelial surface, causing them to slowly roll along the vascular wall.

(2) **Activation**: Endothelial and local cytokine signals induce conformational changes in neutrophil **integrins**.

(3) **Adhesion**: High-affinity binding halts the rolling process, firmly arresting the cell on the endothelium.

(4) **Transendothelial Migration**: Neutrophils move between endothelial cells to enter the inflamed tissue, directed by local cytokine gradients, where they will subsequently phagocytize pathogens.

3.2 Innate Pathogen Pattern Recognition

Core Concept 3.5: Pathogen Associated Molecular Patterns (PAMPs)

Pathogen associated molecular patterns (PAMPs) are conserved molecular structures shared broadly across classes of microorganisms and are not found on host cells, allowing the **innate** immune system to recognize them as generic markers of infection via dedicated **pattern recognition receptors** (*Core Concept 3.6*).

(a) **Proteins** (***very few recognized innately*)
- Flagellin: The primary structural component of bacterial flagella (*tails*).
- Profilin: A surface protein found in protozoa that facilitates their motility and host cell targeting.

(b) **Carbs & Glycopeptides**
- Zymosan: A carbohydrate complex present in fungal cell walls.

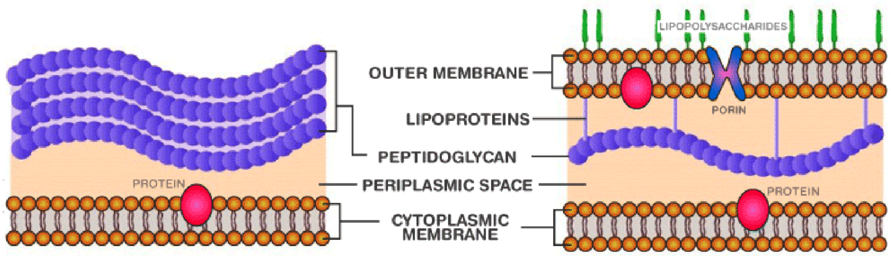

Figure 10: Gram-positive (*left*) and Gram-negative (*right*) bacteria; *peptidoglycan (burgandy) is present in both cell wall types*

- Peptidoglycan: A major structural polymer of both Gram-positive and Gram-negative bacterial cell walls shown in Figure 10.

(c) **Lipids**
- Lipopolysaccharide (LPS): Innately recognized lipids that are typically conjugated to carbohydrates or peptides, exemplified by the outer membrane component of Gram-negative bacteria (*Figure 10*).

(d) **Nucleic Acids**
- Atypical Nucleic Acids: Nucleic acids displaying configurations atypical of the host, such as single-stranded DNA or double-stranded RNA.

Core Concept 3.6: Pattern Recognition Receptors (PRRs)

Pattern recognition receptors (PRRs) are host receptors that recognize PAMPs (*Core Concept 3.5*), enabling the **innate** immune system to detect and respond to a broad range of microbial threats independent of prior antigen exposure.

(a) **Extracellular PRRs**
- Lysozyme: An enzyme that specifically **degrades the peptidoglycan** layer of bacterial cell walls (*Figure 10*).
- Psoriasin: A skin-specific **antimicrobial peptide (AMP)** that **disrupts bacterial cell membranes**; *whereas AMPs are generally distributed systemically and share this membrane-disruptive mechanism, psoriasin is notable for its restriction to cutaneous tissue*.
- Mannose-binding lectin (MBL): A plasma protein that binds microbial carbohydrates to initiate the **lectin pathway of the complement system** (*AUTOREF*).

- C-reactive protein (CRP): An acute-phase plasma protein that binds to specific surface molecules on **microbes** and **damaged host cells**.
- Lipopolysaccharide-binding protein (LBP): A protein that **recognizes and binds lipopolysaccharide (LPS)** (*Figure 10*), the principal constituent of the Gram-negative bacterial outer membrane.

(b) **Cytoplasmic PRRs**

- NOD proteins: Intracellular sensors that detect degradation products of **bacterial cell walls** within the cytoplasm.

(c) **Toll-like receptors (TLRs)** function as monomers or dimers and are distributed either **internally**, on endosomal or phagolysosomal membranes, or **externally**, on the plasma membrane facing the ECM.

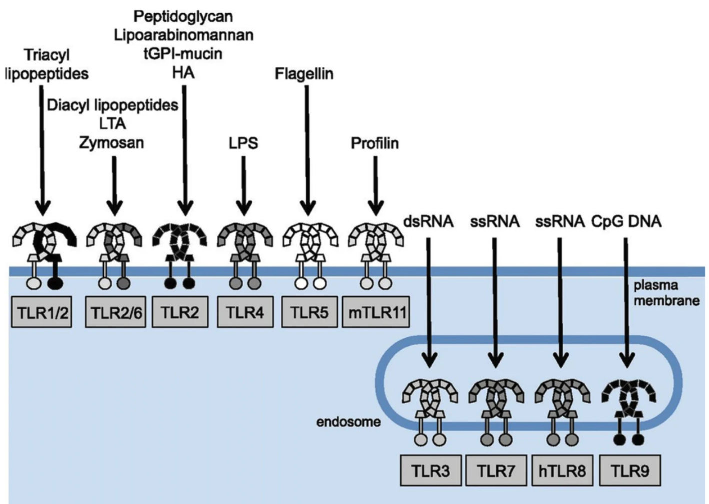

Figure 11: Toll-like receptors; *external (top left) TLRs sit on the plasma membrane facing the ECM and internal (bottom right) TLRs sit on the endosomal membrane*

Upon ligand binding, TLRs initiate signaling cascades that activate the transcription factor NF-κB, upregulating the **production of pro-inflammatory cytokines**.

Category	TLR	Ligand Recognition
External	TLR1, TLR2, TLR6	Mediate the recognition of various **bacterial lipopeptides** and cell wall **lipoproteins**.
	TLR4	Specifically detects **LPS** from Gram-negative bacteria.
	TLR5	Specifically recognizes **bacterial flagellin**.
	TLR10	*Unknown.*
Internal	TLR3	Detects **viral double-stranded RNA**.
	TLR7, TLR8	Detect **single-stranded viral RNA sequences**.
	TLR9	Recognizes **unmethylated CpG dinucleotide motifs** commonly found in bacterial and viral DNA.
Non-Human	TLR11, TLR12, TLR13	These receptors are absent in humans but are expressed in mice, functioning primarily to recognize **profilin** (*an actin-binding protein*) and bacterial 23S rRNA.

Table 3: Ligand specificity of TLRs, categorized by cellular localization.

3.3 Phagocytosis & NOX Complexes

Core Concept 3.7: Phagocytosis

Phagocytosis refers to the engulfment of pathogens via the extension of **pseudopodia** around the target, leading to the formation a a **phagosome**. This process is triggered upstream by **PAMPs** (*Core Concept 3.5*) and their recognition by **PRRs** (*Core Concept 3.6*) and primarily carried out by macrophages and neutrophils, which degrades the trapped pathogen via **NOX complexes** (*Core Concept 3.8*).

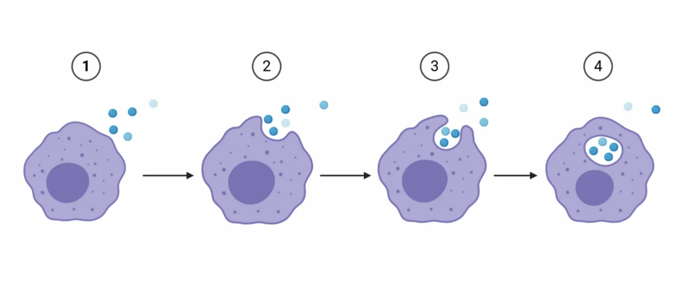

Figure 12: Phagocytosis

Core Concept 3.8: NADPH Oxidase (NOX) Complexes

NADPH oxidase (NOX) complexes are multi-subunit, membrane-bound enzymes whose primary biological function is the targeted generation of **reactive oxygen species (ROS)** through the transfer of electrons across biological membranes.

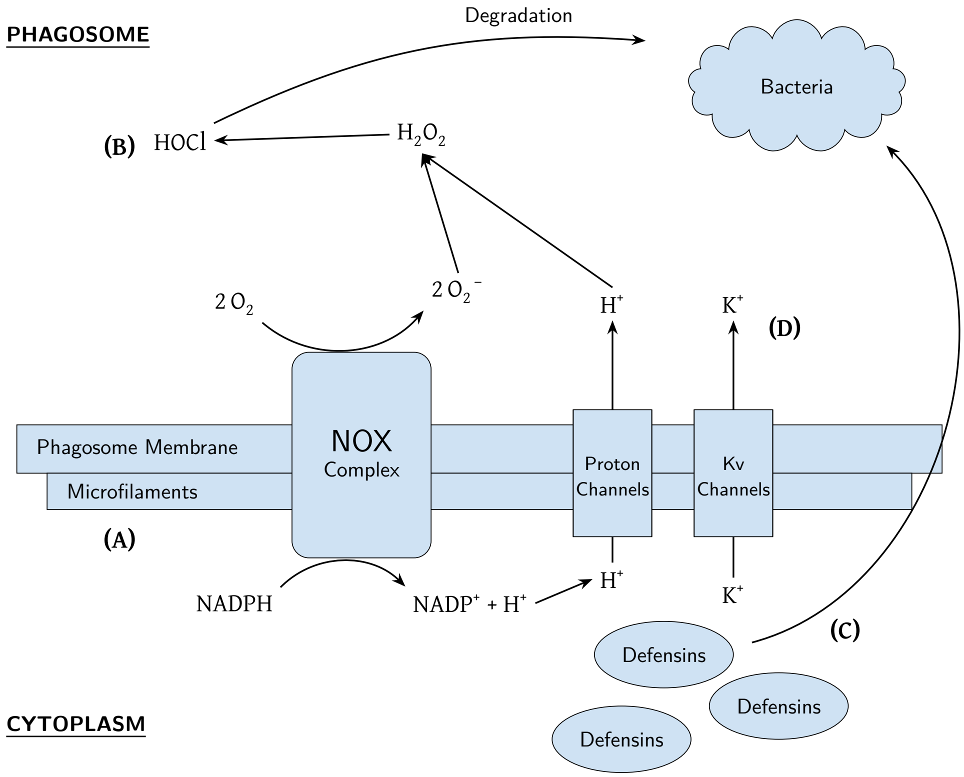

Figure 13: NADPH oxidase (NOX) complex mechanism; *(A)* ***Microfilaments package the exterior*** *of the vacuole, preventing swelling and maintaining the concentration of toxic compounds inside; (B) Both* ***ROS*** *and* ***RNS*** *are made to attack the bacterial cell, this figure only shows the formation of ROS however for simplicity; (C)* ***Defensin peptides (a type of AMP)*** *attack the pathogen by poking holes in its membrane, working best in a certain* K^+ *concentration and tonicity; (D) In order to* ***regulate the electrochemical gradient*** *made by transferring ions across the membrane,* K^+ *is transferred into the phagosomal lumen by membrane channels*

4 Antibodies

4.1 Immunoglobulin Structure & Isotypes

Core Concept 4.1: Immunoglobulin Structure

<FIGURE>

(a) **Light Domain**

- *Variable Region (V_L)*

  Contains three distinct loops extending from the terminus that form the hypervariable region. This region is constructed from three non-contiguous amino acid sequences, comprising approximately 15 to 20 percent of the domain.

- *Constant Region (C_L)*

  The carboxy-terminal portion of this region links with the heavy chain, contributing to the hinge structure that forms the base of the characteristic Y-shape.

  There are two distinct isotopic variants of the light chain constant region:

  Kappa (κ) chains.

  Lambda (λ) chains, which are further divided into five structural subclasses.

(b) **Heavy Domain**

- *Variable Region (V_H)*

  Three terminal loops form the hypervariable region of the heavy chain. When paired spatially with the three corresponding loops of the light chain, they collectively form the complementarity-determining regions (CDRs) responsible for antigen binding.

- *First Constant Domain*

- *Hinge-Bend Region*

  In IgM and IgE (possessing μ and ε heavy chains, respectively), this region consists of additional, highly rigid immunoglobulin domains.

  In IgG, IgA, and IgD (possessing γ, α, and δ heavy chains), this region consists of a proline-rich amino acid sequence stabilized by disulfide bonds, providing substantial structural flexibility.

- *Next Constant Domain*

  This domain serves as the primary site for post-translational oligosaccharide attachment.

  Glycosylation spatially opens the domains to the aqueous environment, essential for facilitating subsequent interactions with complement proteins.

- *Carboxy-Terminal (C-term) Domain*

  This terminal domain dictates the antibody's final localization: whether it is retained as a membrane-bound receptor (on naive or memory B cells) or secreted extracellularly (by plasma cells).

  The secreted variant terminates in a short, exclusively hydrophilic amino acid sequence.

  The membrane-bound variant terminates in a specialized sequence containing a hydrophobic transmembrane span followed by a short cytoplasmic hydrophilic tail.

Core Concept 4.2: Immunoglobulin Isotypes (*H Chain Differences*)

(a) **IgD** (*Memory & Naive B Cells*)

<FIGURE>

Flexible hinge structure.

Functionally assists in initial antigen recognition by naive B cells.

It predominantly operates as a membrane-bound B cell receptor (BCR) and is detected in plasma only at trace levels.

(b) **IgG** (*Memory B Cells*)

<FIGURE>

Flexible hinge structure.

Represents the predominant secreted antibody class responsible for widespread systemic defense against both bacterial and viral pathogens.

Highly effective at activating the complement system, which leads to pathogen opsonization and consequently enhances phagocytic clearance.

<TABLE FOR THE CLASSES>

(c) **IgA** (*Memory B Cells*)

<FIGURE>

Flexible hinge structure.

Primary role involves immune surveillance and defense at mucosal and epithelial barriers.

Circulates in the serum largely as a monomer, but is assembled and secreted across epithelia primarily as a dimer.

The dimeric form is covalently linked by a J (joining) chain, which facilitates secretion into mucus layers and localizes the antibody to various epithelial linings.

The active transport mechanism utilized to move these antibodies across epithelial cell layers is termed transcytosis.

<TABLE FOR THE CLASSES>

(d) **IgM** (*Memory & Naive B Cells*)

<FIGURE>

Rigid hinge region (extra domain).

Serves as a critical first-line defense and is the first antibody isotype secreted into plasma during a primary immune response.

When expressed as a membrane-bound BCR, it exists strictly as a monomer.

The secreted soluble form assembles into a large pentameric structure, possessing ten distinct antigen-binding sites.

The pentamer is stabilized by a central J chain, which aids in mucosal secretion. The multimeric geometry makes IgM highly efficient at cross-linking antigens and initiating the complement cascade.

Like IgG, its potent complement activation results in the rapid opsonization of targets, promoting their phagocytic destruction.

Though larger, it can also undergo transcytosis across mucosal barriers utilizing a similar active transport mechanism.

(e) **IgE** (*Memory B Cells*)

<FIGURE>

Rigid hinge region (extra domain).

Physiologically adapted for immunological defense against large parasitic helminths (worms).

It is deeply implicated in allergic pathogenesis; the constant regions of IgE bind with exceptionally high affinity to specialized Fc receptors on the surface of mast cells and basophils.

4.2 B Cell Receptors

Core Concept 4.3: B Cell Receptors

When an immunoglobulin molecule is anchored in the cellular membrane and oriented outward, it functions as an Immunoglobulin Receptor or B Cell Receptor (BCR).

Prior to activation, naive B cells exclusively co-express IgM and IgD as their surface receptors.

Following activation and differentiation, memory B cells display receptors of other heavy chain classes (e.g., IgG, IgA, IgE). Once a B cell undergoes class-switch recombination to a downstream isotype, that specific isotype becomes the permanent class of receptor it expresses.

<FIGURE>

4.3 Immunoglobulin: DNA to Expression/Secretion

Core Concept 4.4: Immunoglobulin: DNA to Expression/Secretion

<FIGURE>

4.4 Immunoglobulin Genetics

Because somatic cells are diploid, they possess two alleles for every relevant chromosome. Consequently, each genetic rearrangement event has the potential to occur on either allele within the genome. The genetic loci encoding immunoglobulins consist largely of tandemly repeated, homologous segments that function as interchangeable coding blocks during somatic recombination.

Core Concept 4.5: L Chain Gene Sequences

(a) **λ L Chain**

<FIGURE>

Localized to Chromosome 22. V Segments: Approximately 30 to 40 distinct coding blocks. J Segments: 4 distinct coding blocks. C Segments: 5 blocks total, containing 2 functional exons (Cλ1, Cλ2) and 3 pseudogenes (Cλ3, Cλ4, Cλ5).

(b) **κ L Chain**

<FIGURE>

Localized to Chromosome 2. V Segments: Approximately 35 to 40 distinct coding blocks. J Segments: 5 distinct coding blocks. C Segments: 1 functional coding block (Cκ).

<FIGURE>

Further molecular details concerning these sequence loci will be elaborated upon below.

Core Concept 4.6: H Chain Gene Sequences

<FIGURE>

Localized to Chromosome 14.

V Segments: Approximately 40 functional segments.

D Segments: Approximately 23 functional segments.

J Segments: Approximately 6 functional segments.

C Segments: 8 distinct domains that correspond directly to the various antibody isotypes (noting that the closely linked Cμ and Cδ segments are functionally transcribed together prior to alternative splicing).

The specific downstream genomic arrangement of these constant regions follows the order: Cμ/Cδ, Cγ3, Cγ1, Cα1, Cγ2, Cγ4, Cε, and Cα2.

<FIGURE>

Examining the heavy chain constant regions at a higher resolution reveals an organized substructure of coding sequences:

Each region contains 3 to 4 independent exons that correspond to the individual globular domains comprising the structural heavy chain of that isotype.

Situated downstream of the primary structural exons are two highly conserved sequences coding for the hydrophobic membrane-spanning extensions.

Importantly, separate polyadenylation (poly-A) signal sites exist after both the secreted-sequence exons and the membrane-spanning exons, dictating through alternative RNA processing whether the final synthesized antibody will be secreted or membrane-anchored.

4.4.1 L/H Chain Gene Expression

Core Concept 4.7: λ L Chain Gene Expression

(1) **REARRANGEMENT** [Rearranged Gene] — *Nucleus*

<FIGURE>

A single variable (V) segment undergoes somatic recombination with a single joining (J) segment. This DNA rearrangement physically excises any intervening genetic sequences and functionally activates only the promoter immediately upstream of the successfully joined V segment.

<FIGURE>

This process relies on the Recombination Signal Sequence (RSS) mechanism detailed in a subsequent section.

(2) **TRANSCRIPTION** [Primary Transcript] — *Nucleus*

<FIGURE>

RNA polymerase generates a primary transcript covering the rearranged VJ sequence, continuing through the downstream constant (C) region until it terminates at the designated transcription stop signal.

(3) **PROCESSING** [Processed Message] — *Nucleus*

<FIGURE>

Spliceosomes process the primary RNA, removing non-coding intronic sequences located between the rearranged VJ unit and the C domain.

Post-transcriptional modification includes the addition of a poly-A tail at the 3' end to ensure messenger RNA stability.

(4) **TRANSLATION** [Nascent Peptide] — *Into the RER*

<FIGURE>

Ribosomal complexes bind the mature mRNA in the cytosol. Upon synthesizing the initial targeting sequence, the ribosome docks to the Rough Endoplasmic Reticulum (RER), allowing the nascent polypeptide chain to be translocated directly into the ER lumen.

This docking is mediated by a hydrophobic leader (L) sequence at the N-terminus of the developing protein, which interacts with signal recognition particles.

(5) **L REGION SNIPPING** [Mature Chain] — *ER Lumen*

<FIGURE>

Signal peptidases within the ER lumen cleave the N-terminal leader sequence, producing a mature light chain polypeptide consisting solely of the variable (VJ) domain and the constant (C) domain.

Core Concept 4.8: κ L Chain Gene Expression

(1) **REARRANGEMENT** [Rearranged Gene] — *Nucleus*

<FIGURE>

Similar to the lambda chain, a single V segment undergoes recombination with a single J segment. Any intervening J sequences are permanently excised from the DNA, and the upstream promoter corresponding exclusively to the chosen V segment becomes functionally active.

<FIGURE>

This DNA rearrangement is mediated by the RSS molecular mechanism.

(2) **TRANSCRIPTION** [Primary Transcript] — *Nucleus*

<FIGURE>

Transcription initiates at the active promoter, generating a primary RNA transcript that spans the chosen V segment, the selected J segment, any remaining downstream unselected J segments, and the single Cκ constant region.

(3) **PROCESSING** [Processed Message] — *Nucleus*

<FIGURE>

RNA splicing enzymes excise all intronic sequences and the unselected downstream J segments, yielding a contiguous coding message.

A poly-A tail is appended to stabilize the transcript for cytosolic export.

(4) **TRANSLATION** [Nascent Peptide] — *Into the RER*

<FIGURE>

This process proceeds identically to lambda light chain translation, involving RER docking.

(5) **L REGION SNIPPING** [Mature Chain] — *ER Lumen*

<FIGURE>

Proteolytic removal of the leader sequence within the ER yields the final mature protein, comprising the variable and constant regions.

Core Concept 4.9: H Chain Gene Expression

(1) **REARRANGEMENT** [Rearranged Gene] — *Nucleus*

<FIGURE>

The initial recombination event joins one Diversity (D) segment to one J segment. All intervening D and J segments are excised from the genome, while those situated downstream remain intact.

Following the D-J joining, a second recombination event splices a V segment to the newly formed D-J complex. This physically deletes all unselected V segments located downstream of the chosen V segment, and any D segments located upstream of the chosen D segment.

<FIGURE>

<FIGURE>

(2) **TRANSCRIPTION** [Primary Transcript] — *Nucleus*

<FIGURE>

The primary transcript synthesized spans the fully rearranged V-D-J sequence, encompassing all downstream introns and remaining J segments. Importantly, transcription continues unabated through both the Cμ and Cδ constant region coding domains, terminating only after the poly-A site situated downstream of Cδ.

(3) **PROCESSING** [Processed Message] — *Nucleus*

<FIGURE>

The elongated primary transcript is subject to alternative RNA splicing, a critical regulatory step determining whether the mature mRNA will ultimately incorporate the exons for the μ heavy chain (forming IgM) or the δ heavy chain (forming IgD), but not both simultaneously on the same transcript.

Concurrently, all intronic sequences and superfluous unselected J segments are spliced out.

(4) **TRANSLATION** [Nascent Peptide] — *Into the RER*

<FIGURE>

Follows the same ER-docking and luminal translocation process described for light chains.

(5) **L REGION SNIPPING** [Mature Chain] — *ER Lumen*

<FIGURE>

Proteolytic removal of the leader sequence yields the mature heavy chain polypeptide, featuring the highly variable V-D-J domain and the selected constant isotype domain.

Core Concept 4.10: RSS Joining Mechanism

(a) **Juxtaposition**

- *Locating & Cleaving*

  Target loci destined for rearrangement physically relocate to transcriptionally active zones within the nucleus. The chromatin structure relaxes, displacing nucleosomes to grant enzymatic access to the naked DNA.

  The RAG-1/RAG-2 recombinase complex aligns specific Recombination Signal Sequences (RSS), strictly adhering to the "12/23 rule," where a sequence featuring a one-turn spacer must pair with a sequence featuring a two-turn spacer. Upon alignment, an endonucleolytic nick is introduced precisely at the 5' border of the conserved heptamer sequence.

  <FIGURE>

- *Hairpin Formation*

  The newly liberated 3' hydroxyl group executes a nucleophilic attack on the complementary phosphodiester bond of the intact strand. This transesterification reaction creates a covalently sealed hairpin loop at the coding ends of the participating V and J segments (in light chains), or the V, D, and J segments (in heavy chains).

  Conversely, the excised signal ends—containing the heptamer, spacer, and nonamer sequences—are left as a blunt, flush cut featuring a free 5' phosphate group.

  <FIGURE>

(b) **Ligation**

- *Initial Cleavage & Ligation*

  <FIGURE>

  The Artemis nuclease complex resolves the coding hairpins through an asymmetric, random single-stranded cleavage. This imprecise nicking generates varying lengths of unmatched single-stranded DNA overhangs at the joining interface.

- *P-Nucleotide Addition*

  <FIGURE>

  Cellular repair polymerases fill in these asymmetrical overhangs, synthesizing short palindromic (P) nucleotide sequences that conform perfectly to the templates created by the offset hairpin resolution.

- *N-Nucleotide Addition (Heavy Chains Only)*

  <FIGURE>

  In developing heavy chains, the enzyme Terminal Deoxynucleotidyl Transferase (TdT) can independently append up to 15 non-templated (N) nucleotides randomly into the junctional gap prior to final ligation, profoundly expanding the sequence diversity of the resulting antigen receptor.

(c) **Result**

The culmination of this process yields a fully repaired and ligated coding joint, forming the functional antigen-binding domain. Meanwhile, the excised signal sequence fragment—typically forming a closed episomal circle consisting of the joined heptamer-spacer-nonamer elements—is generally degraded, unless specific structural topologies dictated an inversional joining event.

Subsequent to baseline V(D)J recombination, mature B cells undergo Somatic Hypermutation. During active immune responses, point mutations are intentionally introduced into the hypervariable gene loci, providing the substrate for affinity maturation.

It is essential to delineate that somatic hypermutation occurs exclusively in peripheral secondary lymphoid tissues following antigen stimulation, well after the initial B cell maturation phase in the bone marrow has concluded.

4.5 Other

Core Concept 4.11: History: *Edelman & Porter*

In foundational experiments conducted during the 1960s, researchers utilizing serum protein electrophoresis fractionated human plasma into five distinct molecular cohorts. One specific migratory band, designated the "gamma" globulin fraction, was identified as housing the vast majority of the plasma's protective immunological activity. Subsequent purification of this fraction led to the biochemical isolation and characterization of antibody molecules.

<TABLE>

NOTE: Structurally divalent F(ab')2 fragments remain capable of precipitating or cross-linking antigens to form extended immune complexes, whereas monomeric Fab fragments lack this cross-linking ability.

Core Concept 4.12: Haptens (*Small Antigens*)

Haptens are low-molecular-weight chemical entities that can be recognized by antibodies but are inherently non-immunogenic; their minute size prevents them from independently eliciting an adaptive immune response.

To overcome this limitation for experimental or clinical antibody production, haptens must be covalently conjugated to a high-molecular-weight carrier protein, such as bovine serum albumin (BSA), to provide the requisite steric bulk and T cell epitopes needed to induce active immunity.

<FIGURE>

Core Concept 4.13: Monoclonal Antibodies (mAb)

The ability to engineer continuous, uniform production of antibodies featuring customized, highly specific CDR loops targeting defined antigen epitopes holds profound clinical and pharmacological value, especially concerning targeted oncological therapeutics.

Definition: Monoclonal antibodies represent a homogenous population of immunoglobulins synthesized by a single clonal lineage of B cells, resulting in absolute uniformity of paratope structure and antigen specificity.

This production is achieved through the generation of hybridomas—engineered, immortalized cell lines formed by fusing specific primary B cells with myeloma cells, thereby ensuring an indefinite, robust yield of the desired monoclonal antibody construct. The general synthesis methodology is detailed below.

<FIGURE>

5 B Cells

5.1 Naive B Cell Development

Core Concept 5.1: Naive B Cell Development

(1) **Co-Receptor Development**

<FIGURE>

The exposure of Common Lymphoid Progenitors (CLPs) to B-Cell Specific Activator Protein (BSAP) induces the initial surface expression of essential BCR signaling co-receptors, specifically Igα and Igβ, firmly committing the precursor to the B lymphocyte developmental trajectory.

At this stage of commitment, the developing cell is classified as a progenitor B (pro-B) cell.

BSAP expression is maintained constantly across the entire B cell lineage continuum, ceasing only upon terminal differentiation into antibody-secreting plasma cells.

(2) **Heavy Chain**

<FIGURE>

The pro-B cell initiates V(D)J rearrangement at the heavy chain locus. The cell is afforded a maximum of two opportunities (one per allele) to generate a productive rearrangement before default apoptosis occurs.

Upon translating a viable heavy chain polypeptide, the nascent heavy chain associates with a pre-existing surrogate light chain complex, advancing the cell to the precursor B (pre-B) cell stage.

The functional assembly of this surrogate receptor complex delivers a crucial intracellular signal that enforces allelic exclusion at the heavy chain locus, maintains structural stability, and signals the nucleus to begin rearranging the endogenous light chain genes.

(3) **Light Chain**

<FIGURE>

The pre-B cell subsequently attempts V-J rearrangement of the light chain loci, possessing up to four total opportunities (spanning both the κ and λ alleles) before risking apoptosis.

A productive rearrangement yields a functional light chain that pairs with the heavy chain, resulting in the surface expression of a complete IgM BCR featuring a fully defined complementarity-determining region (CDR).

The cell, now bearing its unique surface receptor, transitions into the immature B cell designation.

(4) **Negative Selection**

Within the bone marrow microenvironment, immature B cells undergo negative selection to rigorously test their newly formed BCRs for potentially harmful autoreactivity against local self-antigens.

Approximately 10% of the initial developing B cell pool successfully navigates this strict tolerance checkpoint.

Autoreactive cells face elimination; binding of widespread multivalent self-antigens heavily crosslinks the surface IgM, triggering rapid apoptotic cell death pathways.

However, under specific conditions, an autoreactive B cell may avoid immediate destruction by reactivating Recombination Activating Gene (RAG) transcription, initiating a process termed receptor editing to generate a new, non-autoreactive light chain specificity.

Cells that successfully bypass negative selection exhibit functional central tolerance, lacking recognition for endogenous self-antigens.

(5) **IgD Formation**

<FIGURE>

Surviving cells employ alternative mRNA splicing to concurrently transcribe and translate the IgD isotype alongside their existing IgM receptors.

The dual expression of surface IgM and IgD signifies a fully mature, immunocompetent, yet naive B cell, which is now authorized to exit the bone marrow and enter systemic circulation.

(6) **During Circulation**

The mature B cells migrate through the circulatory and lymphatic systems to seed peripheral secondary lymphoid organs, including the lymph nodes, spleen, and various mucosal tissues.

Upon encountering specific antigen and receiving activating signals, the resulting plasma cells drastically upregulate transcription of the μ heavy chain, altering RNA processing to exclude the transmembrane coding domains, leading to massive secretion of the soluble, multimeric IgM pentamer.

Soluble secretion of the δ isotype (IgD) is physiologically negligible, as the cell rarely produces δ transcripts lacking the membrane anchor sequences.

5.2 B Cell Activation and Proliferation

5.2.1 Thymus Independent and Dependent B Cell Activation

Core Concept 5.2: Thymus Independent B Cell Activation

Certain distinct classes of antigens, termed Thymus-Independent (TI) antigens, possess the unique capability to induce B cell activation and proliferation in the complete absence of concurrent T helper (TH) cell stimulation. These structures frequently co-engage innate Toll-Like Receptors (TLRs).

TI-mediated activation is characterized by two fundamental limitations: it generally fails to induce functional immunoglobulin class switching (restricting output predominantly to IgM), and it does not yield a sustained population of memory B cells; robust induction of either process strictly requires classical TH cell involvement.

(a) **Type 1**

<FIGURE>

TI-1 antigens typically consist of potent mitogenic molecules, most notably lipopolysaccharide (LPS), derived from the outer leaflet of Gram-negative bacterial cell walls.

At significant concentrations, these molecules bind the BCR while simultaneously acting as potent ligands for adjacent TLR4 receptors, delivering the necessary, self-contained secondary permissive signal for cellular activation.

(b) **Type 2**

<FIGURE>

TI-2 antigens are characterized by highly repetitive, polymeric structural motifs (such as extensive bacterial capsular polysaccharides or flagellin polymers) that induce massive, sustained cross-linking and clustering of surface BCRs.

While lacking direct TH cell contact, successful TI-2 activation often relies upon a generalized, supportive microenvironment heavily supplemented by inflammatory cytokines (e.g., IL-2, IL-3, and IFN-γ) secreted by nearby innate immune populations and specialized APCs.

Core Concept 5.3: Thymus Dependent B Cell Activation

Under canonical circumstances governing the majority of adaptive responses, specific antigen binding heavily cross-links the B cell's surface immunoglobulin. This initial binding event triggers an intracellular cascade that structurally primes the B cell and directs it to actively solicit requisite co-stimulation from an antigen-specific TH cell.

<FIGURE>

(1) Extracellular antigen physically cross-links multiple BCR complexes. This physical clustering forces the spatial aggregation of the associated intracellular Igα/Igβ signaling domains, leading to the rapid structural activation of their Immunoreceptor Tyrosine-based Activation Motifs (ITAMs).

(2) Resident Src-family kinases bind these aggregated ITAMs, utilizing ATP hydrolysis to systematically phosphorylate critical tyrosine residues on the motifs.

(3) The newly phosphorylated ITAMs serve as high-affinity docking sites for Syk kinase. The recruitment and activation of Syk subsequently initiates a vast, highly interconnected intracellular signaling web:

    (a) DAG/IP3 Generation: Phospholipase C-γ (PLC-γ) is activated, cleaving the membrane phospholipid PIP2 into diacylglycerol (DAG) and inositol trisphosphate (IP3), acting as pivotal second messengers.

    IP3 diffuses to trigger the rapid release of stored Ca2+ from the ER lumen into the cytosol. This calcium influx heavily binds calmodulin, which subsequently activates the phosphatase calcineurin to dephosphorylate and activate the transcription factor NFAT.

Concurrently, DAG activation propagates signals that result in the targeted degradation of the IκB inhibitory complex, thereby unleashing the potent pro-inflammatory transcription factor NF-κB to enter the nucleus.

(b) MAPK/ERK Kinase Cascade: Through intermediary adapter proteins, the classical RAS-RAF-MEK-ERK phosphorylation cascade is initiated. This culminates in the nuclear translocation of activated ERK proteins, which directly upregulate essential target genes controlling robust cellular proliferation and terminal differentiation.

(c) PI3K Activation: The activated BCR complex recruits PI3K, which phosphorylates residual membrane PIP2 to yield PIP3. The accumulation of PIP3 provides a powerful overarching survival signal, effectively blocking endogenous apoptotic pathways during the stress of rapid clonal expansion.

(4) Driven by these integrated transcription factors, the fully activated B cells commence exponential mitotic division and begin differentiating into functional, antibody-secreting plasma cells.

(5) Feedback Regulation: As the immune response progresses, the accumulating high-titer antigen-antibody immune complexes begin to bind inhibitory surface Fcγ receptors, specifically CD32 (FcγRIIB) which houses an ITIM domain, providing essential negative feedback to safely curtail the response. (Note: CD22 is an independent inhibitory coreceptor recognizing sialic acid, playing a similar dampening role).

(a) While strictly necessary for controlling inflammation, suppressing active signaling presents a risk of inducing peripheral tolerance toward foreign antigens—a state where the immune system inappropriately acknowledges a foreign entity without mounting a defense. This is heavily counterproductive during active pathogenic infections.

(b) To carefully manage this balance, specialized Regulatory T (Treg) cells act primarily to enforce systemic tolerance specifically directed against autologous host proteins, as well as maintaining non-reactivity toward benign, commensal microbiome flora.

(c) Clinically, if exogenous antibodies targeting a specific antigen are passively administered into a host system, they will effectively sequester the circulating antigen load, largely preventing it from natively binding host BCRs and thus actively suppressing the generation of a de novo immune response.

Core Concept 5.4: T_H Cell Synpase

Despite robust BCR engagement, sustained proliferation heavily relies upon physical contact with an activated TH cell. Notably, the high affinity of the BCR allows the B cell to capture and present antigen to TH cells at concentrations roughly 100 to 10,000 times lower than what is required for standard presentation by macrophages or dendritic cells.

(1) BCR engagement triggers receptor-mediated endocytosis, allowing the B cell to internalize the bound antigen and subsequently hydrolyze it into small peptide fragments via the endosomal pathway.

(2) These processed antigenic peptides are then loaded onto nascent MHC class II molecules and shuttled back to the external cell surface for presentation.

(3) An activated TH cell specific for that exact peptide epitope binds the MHC II complex via its TCR, heavily stabilized by the CD4 co-receptor, forming a highly organized cellular interface termed the immunological synapse.

(4) This direct recognition event induces the TH cell to rapidly upregulate surface expression of CD40 Ligand (CD40L/CD154), which securely engages the constitutively expressed CD40 receptor physically present on the B cell surface.

(5) The CD40-CD40L interaction delivers the critical, definitive second signal to the B cell. This massively

amplifies the internal signaling cascades initiated by the BCR, fully committing the B cell to exponential proliferation and terminal effector differentiation.

In this reciprocal exchange, the B cell concurrently upregulates surface B7 molecules (CD80/CD86), which engage CD28 on the TH cell to continually support and reinforce the TH cell's activated state.

5.2.2 Antibody Development

Core Concept 5.5: Lymph Node Follicle Development

<FIGURE>

Circulating free antigen and small immune complexes present in the plasma or lymph are predominantly captured by local antigen-presenting cells (APCs).

These APCs traffic to the lymph node to initially activate TH cells within the paracortical region. These primed TH cells then migrate toward the B cell zones, forming functional immunological synapses with antigen-bearing B cells. This interaction generates a proliferating cellular cluster localized precisely at the boundary delineating the paracortex and the outer cortex.

This active cluster rapidly expands and reorganizes, driving the transformation of a resting primary follicle into a highly active secondary follicle, distinctly characterized by a well-defined light germinal center and a dense dark zone.

Germinal Center: The primary structural environment facilitating the intense interactions among B cells, specialized TH cells, and FDCs required for rigorous antibody selection.

Dark Zone: The anatomical location dedicated to extreme B cell proliferation accompanied by targeted somatic hypermutation.

NOTE: Within this environment, FDCs trap and stabilize intact antigen-antibody immune complexes within distinct beaded cellular structures termed iccosomes, presenting them efficiently for selection by competing B cell clones.

Core Concept 5.6: Affinity Maturation

<FIGURE>

Centroblasts—highly activated, morphologically enlarged B cells—migrate into the dark zone where they experience intense mitotic proliferation. Concurrently, they undergo somatic hypermutation, introducing targeted nucleotide substitutions strictly within the DNA sequences coding for the CDR loops. Throughout this proliferative stage, centroblasts temporarily cease the surface expression of their BCRs.

Upon exiting the cell cycle, these cells mature into centrocytes and migrate toward the light zone for a rigorous selection process. Centrocytes must actively compete to bind the limited antigen presented on FDC iccosomes. Cells possessing mutated receptors with improved affinity successfully bind the antigen, enabling them to present peptide to TH cells, thereby securing the vital CD40 survival signals. This process functions essentially as accelerated, localized natural selection.

Centrocytes bearing low-affinity or non-functional mutations fail to secure sufficient antigen or TH cell help; consequently, they undergo rapid apoptosis and are cleared locally by tingible body macrophages.

The highly selected, high-affinity centrocytes finally exit the germinal center microenvironment, terminally differentiating into long-lived plasma cells. These cells permanently downregulate surface receptor expression in favor of massive, sustained secretion of high-affinity antibodies targeted against the infection.

Core Concept 5.7: Class Switching

TH cells maintain synaptic contact with differentiating B cells, secreting tailored cytokine profiles that chemically direct class-switch recombination. This mechanism physically targets specific ”switch regions”—conserved sequences spanning 2 to 3 kilobases positioned immediately upstream of each respective constant gene segment.

Prior to any switching event, the mature B cell exclusively transcribes the proximal μ and δ constant regions, forming its initial IgM/IgD receptors.

The switching mechanism involves the permanent excisional recombination of DNA, physically looping out and discarding the previously utilized constant domains alongside any intervening sequences positioned between the rearranged VDJ domain and the newly targeted downstream constant region.

Because this deletion occurs at the DNA level, any excised upstream constant regions are irretrievably lost from that allele’s genome. Therefore, progressive class switching is unidirectional down the heavy chain locus.

This large-scale genetic remodeling generally requires a sustained immune response, typically occurring functionally prominent after a week or more of persistent antigen challenge.

Consequently, the primary serological hallmark of a novel acute infection is a rapid spike in low-affinity IgM; the transition to high-affinity IgG (or other specialized isotypes) is a slower, distinctly subsequent event characterizing the later phases of the adaptive response.

<FIGURE>

5.2.3 Plasma vs Memory B Cells

Core Concept 5.8: Plasma Cells

The terminal differentiation from an activated centrocyte into a highly specialized plasma cell is orchestrated by key transcription factors. This shifts the cellular machinery, altering RNA splicing enzymes to selectively exclude the membrane-anchoring exons from the heavy chain transcript, thereby converting receptor expression into large-scale soluble secretion.

This terminal transition is strongly correlated with, and facilitated by, the significant downregulation and suppression of BSAP (Pax-5) expression.

Core Concept 5.9: Memory Cells

Conversely, cells diverted from terminal differentiation to become long-lived memory B cells possess a quiescent morphology superficially resembling naive B cells. However, they bear the genetic scars of class switching and affinity maturation, expressing specialized, high-affinity isotypes (such as IgG, IgA, or IgE) rather than the baseline IgM/IgD phenotype restricted to naive cohorts.

The stabilization and maintenance of the memory cell lineage is highly dependent on the continued, elevated expression of the BSAP transcription factor.

5.3 Leukemia & Lymphoma

Core Concept 5.10: Leukemia & Lymphoma

The Burden of Non-Productive Rearrangements

The highly error-prone and structurally randomized nature of the V(D)J recombination process introduces immense genomic instability.

Statistically, the imprecise resolution of coding joints possesses roughly a 66% probability of introducing a catastrophic frameshift mutation during any single recombination event.

Furthermore, the stochastic action of Artemis (P-nucleotides) and TdT (N-nucleotides) frequently generates premature, non-sense stop codons directly within the reading frame.

Failing to generate structurally viable heavy and light chains results in a non-productive rearrangement, inevitably condemning the developing progenitor cell to immediate apoptosis.

Allelic Exhaustion

A developing cell will exhaust all available allelic options—attempting rearrangement on both heavy chain alleles and across the four total light chain alleles—before finally succumbing to apoptotic clearance.

This intense requirement for repeated, double-stranded DNA breaks and rampant cellular proliferation inherently predisposes the B cell lineage to oncogenic transformation. Erroneous recombination events, such as translocations involving the highly active Ig promoters and critical proto-oncogenes, form the fundamental molecular basis for many lymphomas and leukemias.

<FIGURE>

6 Complement System

6.1 MAC Complex Attack

Core Concept 6.1: MAC Complex Attack

<FIGURE>

This process is called membrane lysis (poking holes in pathogenic membranes)

The MAC pathway can be triggered through both innate and adaptive recognition—this is the last stage of the complement activation pathway (all complement pathways converge to the cleavage of C5)

The MAC complex is essentially a ring of C9 peptides that serves as a passage hole for harmful substances to degrade the insides of cells like...

membrane coated viruses

rough gram negative (membrane-peptidoglycan-membrane); gram positive is difficult because of its thick walls

eukaryotic parasites

erythrocytes or RBCs (under pathogenic conditions)

cancer cells

The size of the MAC complex is dependent on the number of C9s that forms the ring (70 to 100)

Most complements are synthesized by the liver and release in an inactive form (VERY SIMPLIFIED)

6.2 Complement Pathways

Core Concept 6.2: Classical Pathway

<FIGURE>

An inactive C1 complex—composed of 6 C1q (C1qA, C1qB, C1qC), 2 C1r, and 2C1s subunits—activates when at least 2 of the C1q subunits' heads bind to antibodies, making the C1rs complex wrap on the C1q stems and exposing the C1rs complex's catalytic activity'

The oligosaccharide and the Ig domain it is attached to constitute the principal agent of activation

Activating Antibody Isotypes

IgM—very effective—usually floats in a planar conformation within plasma, but when binded to an antigen, it protrudes it's Fc stem outwards (making it easier to bind)

IgG is monomeric. The C1 complex needs two IgGs to bind to in order to go down its pathway (this prevents random activation)

IgG3 is the best, IgG1 is the second best, IgG2 barely activates, IgG4 doesn't work at all

IgA, IgE, and IgD do not activate the complement system

<FIGURE>

Core Concept 6.3: Lectin Pathway

The Lectin pathway looks the same as the Classical pathway except these few points

<TABLE>

The rest of the pathway starting from the cleavage of C4 and C2 are the same

Core Concept 6.4: Alternative Pathway

<FIGURE>

Core Concept 6.5: Side Signal Pathways & Other

Intro sentence

(a) **Inflammation** (*Anaphylatoxin*)

- *C3a*: Triggers degranulation of mast cells and basophils, with also permeability changes in the endothelium
- *C5a*: Summons macrophages and neutrophils

(b) **Coating**

- *Opsonization*: Complements bind to pathogenic cells, enhancing phagocytosis towards them—[C3b]
- *Viral Neutralization*: Simply attaching a complement to a host cell invader makes it much more difficult to invade
- *Immune Complex Clearing*: Immune complexes (ICs) like antigen-antibody and antigen-complement complexes tend to bind to RBCs and get filtered as then circulate

7 Major Histocompatibility Complex

7.1 MHC Structure & Genetics

Core Concept 7.1: MHC Characteristics

<FIGURE>

<TABLE>

Core Concept 7.2: MHC Gene Sequences

<FIGURE>

The MHC gene sequences is found in chromosome 6

This whole block of genes is termed a haplotype

MHC in humans is commonly referred to as Human Leukocyte Antigen (HLA) and is part of the immunoglobulin superfamily

REMINDER: there are 2 copies of these gene sequences (one from mom, one from dad)

(a) **MHC I**

Further separated by three different versions: HLA-A, HLA-B, HLA-C (each with a different specificity for peptide binding)

There are currently 68 HLA-A, 125 HLA-B, and 44 HLA-C version within the human population that have been identified (thus an underestimate)

An individual will express at max 6 different types of MHC I proteins—assuming that the maternal alleles are distinct from the paternal alleles

***This gene sequence only covers the α subunits; the β subunit locus is on chromosome 15 (non-polymorphic)

WILDLY POLYMORPHIC (high allele counts for HLA versions)

NO ALTERNATIVE SPLICING FORMS (DNA is not changed)

(b) **MHC III**

This gene sequence is completely different from the main MHC proteins I and II

It is included just because it happens to lie in between

Codes for secreted proteins involved in innate response: [complement proteins], [cytokines], [heat shock proteins]

(c) **MHC II**

Each subsection—HLA-DP, HLA-DQ, HLA-DR (each with a different specificity for peptide binding)—codes for 2 separate peptides (α and β) which function together to make up the MHC II protein

An individual will express at max 12 different types of MHC II proteins (due to potential paternal-maternal α-β dimers)—assuming that the maternal alleles are distinct from the paternal alleles

WILDLY POLYMORPHIC (high allele counts for HLA versions)

NO ALTERNATIVE SPLICING FORMS (DNA is not changed)

(d) **Other**

Interspersed in the MHC II genes are those that code for elements of the antigen processing system for MHC I

There are other MHC I proteins involved in other functions (specialized receptor, transport, presentation)

Core Concept 7.3: MHC Mendalian Genetics

(a) **Non-Recombinant Inheritance**

<FIGURE>

The alleles coding for the MHC gene sequences tend to be passed on to offspring as a block (with no crossing over; tightly linked)—thus, in most cases, siblings will have a ¼ chance to have the same MHC genes

**The term "haplotype" is used in other contexts where a group of genes remains in a block and be passed on as a unit (e.g. & chromosomes and Mitochondrial DNA)

(b) **Recombinant Inheritance**

<FIGURE>

Rarely, if the cross over happens between the I and II block, this produces a recombinant haplotype (highly unlikely to have sibling match)

This increases variability into the gene pool

Inbreeding

With inbreeding—there will be no introduction of new haplotypes of the MHC gene sequences within the population—thus each individual will have the same haplotype on each homologous chromosome

This individual will thus have one 3 versions of MHC I and 3 versions of MHC II rather than the maximum 6 and 12 respectively

7.2 MHC Antigen Preparation

Core Concept 7.4: Endogenous MHC I (T_c Cells)

(1) **Proteasomal Degradation of Antigen to Peptide**

<FIGURE>

The antigen is targeted for degradation by binding to ubiquitin—a small protein—and is sent to the interior of a proteasome

Proteosome is made up of a protein with a 7-fold symmetry (associated with danger) and is more active under conditions of stress and damage by their nature

LMP2 and LMP7 replace proteasomal subunits under inflammatory conditions and promote degradation into the correct peptide size (9 amino acids)

There are also complex backup systems for this

LMP2 and LMP7 genes are located within the MHC II gene sequence and are polymorphic

(2) **Transport to the RER**

<FIGURE>

The TAP1 and TAP2 bind to the peptides, hydrolyze ATP, and drive them into the RER lumen

TAP1 and TAP2 genes are located within the MHC II gene sequence and are polymorphic

TAP Protein Structure

Heterodimer

Cytosolic binding side for peptides

ATP hydrolysis site

Hydrophobic anchoring regions with 3 switchbacks per peptide

(3) **Assembly of MHC I**

(a) *Calnexin-α to α-β*

<FIGURE>

The α chain is inserted into the membrane, and the β microglobulin chain sent through to the lumen

Here calnexin ensures the α chain's proper folding

Once the β microglobulin chain bind to the α chain, calnexin unbinds

(b) *α-β-Calreticulin-Tapasin-TAP-ERp57*

<FIGURE>

The complex associates within calreticulin and tapasin, and then ERp57 binds over the cleft

Calreticulin and ERp57 both help stabilize the molecule and prepare it for peptide binding

Tapasin brings over the TAP complex over to the pre-peptide MHC I

(c) *α-β-Calreticulin-Tapasin-TAP-Peptide*

<FIGURE>

TAP protein hands off the peptide via tapasin to the cleft which displaces ERp57

(d) *α-β-Peptide to Golgi*

<FIGURE>

Calreticulin and tapasin dissociate from the MHC I-peptide complex and travel to the Golgi

(4) **Transportation to Exterior**

<FIGURE>

After final sorting the Golgi, the vesicle with MHC I-peptide complex is released by exocytosis onto the membrane exterior

Core Concept 7.5: Exogenous MHC I / Cross Presentation (T_c Cells)

(1) **Antigen Uptake**

<FIGURE>

Through endocytosis, the cell picks up the exogenous antigen

(2) **Proteasomal Degradation of Antigen to Peptide**

<FIGURE>

After getting degraded within the endosome through degradation stages, the peptide fragments travel to and fuse with the RER, exposing the exogenous peptides to MHC I proteins

NOTE: This process only happens in APCs, and activated TC cells do not attack the APCs because they don't express co-stimulatory danger signals that promote an attack on the presenting cell The purpose of cross presentation is to lower the response time of TC cells.

Core Concept 7.6: Exogenous MHC II

(1) **Antigen Processing**

(a) *Antigen Uptake*

<FIGURE>

The antigen is internalized through phagocytosis—a specialized form of endocytosis during which there occurs an engulfment of whole particles with pathogen

The formed vesicle injkj the cell interior is named an endosome

(b) *Hydrolysis in Vesicles*

<FIGURE>

An endosome goes through three stages (each one more acidic and hydrolytic) and use lysosomes to degrade the foreign antigens (and remove carbohydrates and lipids), cutting them up to their proper size (13-18 amino acids long)

Early endosome, pH 6.0 - 6.5

Late endosomes of endolysosomes, pH 5.0 - 6.0

Lysosomes (or phagolysosomes), pH 4.5 - 5.0

(2) **MHC II**

(a) *Preparation of MHC II*

<FIGURE>

After translation of the MHC II αβ heterodimer onto the RER membrane, the heterodimer associates with the invariant chain (Ii)

The figure only shows one invariant chain binding, but in reality, the Ii comes as a trimer—three MHC II heterodimers bind at a time

The invariant chain [1] blocks the cleft (to prevent immature binding), [2] ensures proper conformation, [3] helps RER exiting, and [4] signals for fusing with phagolysosomes

(b) *Transportation of Golgi*

<FIGURE>

The αβ-Ii complex is then transported to the Golgi where it awaits fusion with a phagolysosome

(3) **Fusion with Endocytic Vesicles/Phagolysosomes**

<FIGURE>

The αβ-Ii complex fuses with early endosomes after leaving the Golgi

(4) **Invariant Chain Degradation**

<FIGURE>

After the development of the endosomes into more acidic internal environments, Ii gets degraded, leaving only the CLIP attached to the MHC II and still blocking the cleft region

(5) **CLIP Removal**

<FIGURE>

At the end of antigen degradation into peptides, non-classical MHC (HLA-DM) removes the CLIP, allowing for peptide binding and the formation of a peptide-presenting MHC II protein (which is very stable)

(6) **Transportation to Exterior**

<FIGURE>

The peptide-MHC II complex is then transported to the cell surface to be displayed

Core Concept 7.7: Exogenous CD1 MHC

(1) **Glycolipid Uptake**

(2) **Preparation and Expression of CD1 MHC**

(3) **Retreat of CD1 MHC**

(4) **Binding of CD1 MHC to Glycolipid**

(5) **Transportation to Exterior**

NOTE: Several different types of cells and receptors can bind to and respond to the CD1 lipid presentation

T cells bind to the lipid-CD1 MHC complex through the γδ receptor (with no CD4 or CD8)

TH cells can also bind through the αβ receptors using CD4

NK cells

invariant natural killer T cells (iNKT) with CD1

8 T Cells

8.1 T Cell Receptor

8.1.1 TCR Complex Characteristics

<FIGURE>

Core Concept 8.1: αβ and γδ Receptors

<FIGURE>

T cells can make either the αβ or γδ TCR—a heterodimer (most T cells make the αβ version)

Both resemble a truncated Ig and is part of the Ig superfamily with three hypervariable loops in each chain (CDRs)

The positively charged transmembrane regions of the TCR and negatively charged transmembrane regions of the CD3 signal transducers form a salt bridge attraction that holds the complex together

During development, the TCR does not go through affinity maturation like B cells, thus they need the supplement receptors (CD4/8) in order to strengthen this less specific binding

αβ Receptor

Undergoes selection to produce a highly specific recognition molecule

γδ Receptor

Undergoes selection to produce a quasi innate recognition molecule (some specificity but not as much as αβ)

Mostly for recognizing lipid antigens of bacteria and parasites

Important in patrolling mucosa and epithelium

Core Concept 8.2: γε and δε Heterodimers

<FIGURE>

All chains (γ, δ, ε) are part of the Ig superfamily

All peptides have an immunoreceptor tyrosine-based activation motif (ITAM) at the cell-interior side of the receptor

Core Concept 8.3: ζζ or ζη Heterodimers

<FIGURE>

Chains are NOT part of the Ig superfamily

Only 9 amino acids to the exterior and a long extended cytoplasmic tail

Has 3 ITAMs rather can one

8.1.2 CD4 & CD8

Core Concept 8.4: CD4

<FIGURE>

Used by TH cells to bind to MHC II

Monomer

4 extracellular Ig domains

N-terminal binds to the base of MHC II (junction of α2 and β2 domains) at a hydrophobic pocket

C-terminal possess 3 serine regions that can be phosphorylated

Core Concept 8.5: CD8

<FIGURE>

Used by TC cells to bind to MHC I

Dimer—α-β hetero or α-α homo linked by disulfide

1 extracellular Ig domain per peptide

N-terminal binds to the α2 and β microglobulin

C-terminal possess amino acid regions that can be phosphorylated

8.1.3 TCR Genes and Rearrangement

Core Concept 8.6: Gene Sequence and Receptor Overview

<FIGURE>

(a) **α & δ Peptide**

<FIGURE>

Chromosome 14 Long Arm

Alpha (α): Resembles the light chain Ig gene because of absence of D regions 40 V — N/A D — 50 J — 1 C

Delta (δ): Resembles the heavy chain Ig gene because of presence of D regions 3 (or more) V — 3 D — 4 J — 2 C

(b) **β Peptide**

<FIGURE>

Chromosome 7 Long Arm

Resembles the heavy chain Ig gene because of presence of D regions 40 V — 2 (functional) D — 6 and 7 J — 2 C

(c) **γ Peptide**

<FIGURE>

Chromosome 7 Short Arm

Resembles the light chain Ig gene because of absence of D regions 5 (functional) V — N/A D — 5 J — 2 C

Core Concept 8.7: α Rearrangement

(1) **Gene Sequence**

<FIGURE>

(2) **δ Removal & VL-J Joining**

<FIGURE>

**Splicing and joining of the sequences are carried out through the same process of RSS manipulation by RAG enzymes done in Ig gene rearrangement ALSO P and N nucleotide addition occurs and further adds to the variability

(3) **Result**

<FIGURE>

[Variable # of VL] - [VL] - [J] - [Variable # of J] - [Intron] - [C]

Next is EXPRESSION

Core Concept 8.8: β Rearrangement

(1) **Gene Sequence**

<FIGURE>

(2) **VL-D/J Joining**

- *VL-DJ*

  <FIGURE>

  VL joins with a D

  **Splicing and joining of the sequences are carried out through the same process of RSS manipulation by RAG enzymes done in Ig gene rearrangement ALSO P and N nucleotide addition occurs and further adds to the variability

- *VL-J*

  <FIGURE>

VL joins with a J (all Ds get cut out)

**Splicing and joining of the sequences are carried out through the same process of RSS manipulation by RAG enzymes done in Ig gene rearrangement ALSO P and N nucleotide addition occurs and further adds to the variability

(3) **Result**

- *VL-DJ*

  <FIGURE>

- *VL-J*

  <FIGURE>

Next occurs EXPRESSION

Core Concept 8.9: γ Rearrangement

(1) **Gene Sequence**

<FIGURE>

(2) **VL-D/J Joining**

<FIGURE>

**Splicing and joining of the sequences are carried out through the same process of RSS manipulation by RAG enzymes done in Ig gene rearrangement ALSO P and N nucleotide addition occurs and further adds to the variability

(3) **Result**

<FIGURE>

Next occurs EXPRESSION

Core Concept 8.10: δ Rearrangement

(1) **Gene Sequence**

<FIGURE>

(2) **VL-D/J Joining & Result**

<FIGURE>

The VL sequence either joins to a D or J depending on what it splices out

NOTE: This rearrangement can add more than one D to the sequence as well

**Splicing and joining of the sequences are carried out through the same process of RSS manipulation by RAG enzymes done in Ig gene rearrangement ALSO P and N nucleotide addition occurs and further adds to the variability

Next occurs EXPRESSION

Core Concept 8.11: Subsequent Expression

(1) The message gets trancribed

NOTE: TCR C regions always has its bulk C gene sequence followed by an extracellular gene (XC), a transmembrane gene (TM), and a cytoplasmic tail gene (CA)

<FIGURE>

(2) The message precursor is capped and gets a poly-A-tail

(3) Processing occurs (removing the introns and other extra Js)

<FIGURE>

(4) Translation of the L region attaches the ribosome to the RER and inserts into the lumen with the C-terminal bound to the ER membrane

(5) The L region is clipped off

(6) Enzymes in the Golgi form disulfide bonds

(7) Vesicles with finished TCRs get transported to and fuses with the plasma membrane

Core Concept 8.12: History: *Hendrick and Davis*

Hendrick and Davis were responsible for the gene identification of TCRs

(1) **Selecting Membrane Bound Polysomes**

<FIGURE>

H&D first separated membrane bound from cytoplasmic proteins by isolating mRNA from membrane bound polysomes

T cells—TCR, CD3, CD4/8

B cells—Ig receptor (BjkjkCR), co-receptors

Both of these cells with have other things in common as well

(2) **Subtractive Hybridization**

<FIGURE>

H&D used membrane bound B cell genes to selectively hybridize with and remove any overlapping radioactive cDNA from T Cells, leaving only the radioactive cDNA unique to T cells behind

(3) **Gene Comparison Across Multiple T Cells**

<FIGURE>

Now there are only 10 different cDNA clones remained (unique to T cells) and we have to find which ones are rearranged (which would be the TCR)

We run each cDNA portion across multiple T cells lineages

If the cDNA portion (for example, g1) is not rearranged, the length traveled in the gel electrophoresis would be the same across all T cell lineages and B cells (control) as well

If the cDNA portion (for example, g2) is rearranged, the length traveled in the gel electrophoresis would be different across different T cell lineages because of their variable lengths

The rearranged gene were 2 out of the 10 cDNAs, thus locating the genes for the TCR

Alloreactivity is T cells being able to bind with foreign MHCs, which it should not be able to do (Figure on MHC Characteristics)

8.2 T Cell Development

Core Concept 8.13: Developmental Pathway

<TABLE>

8.3 TCR Antigen Response

Core Concept 8.14: Antigen Response Pathway

(1) **Antigen Recognition**

APC presents an antigenic peptide on MHC II

(2) **Binding**

TH cell binds to this complex through TCR–CD3 complex, aided by CD4

*Superantigens are viral or bacterial proteins that nail together TCR-antigen-MHC complexes, hyperstimulating the TH cells and overproducing cytokines (which could kill you)

(3) **Signaling Cascade**

A signaling cascade starts towards the nucleus and pushes the cell into cell division

(4) **Transcriptional Activation**

- *Immediate Genes*

  Genes for transcription factors for inflammatory response

  —NFAT, AP-1, NF-κB

- *Early Genes*

  Cytokines, Interleukins and their receptors

  *This is where developmental branching occurs, and the TH cells differentiate into either TH1, TH2, THreg, or TH17 (sometimes referred to as TH3)

  Inducing cytokines are produced by other immune cells like macrophages and DCs which during the production of master transcription factors

  —[Type]: [inducing cytokine] → [master transcription factor]

  —Th1: IL-12 → T-bet

  —Th2: IL-4 → GATA3

  —Th17: TGF-β + IL-6 → RORγt

  —Treg: TGF-β + IL-2 → FOXP3

- *Late Genes*

  Cell adhesion molecules (CAMs) and effector cytokines

Core Concept 8.15: TCR Antigen Recognition to ZAP70

<FIGURE>

(1) APCs present their MHC II and antigen complex to the TCR

(2) CD4 helps strengthen this binding

(3) p56lck (which is associated with CD4/8 on its cytoplasmic tail) phosphorylates the ITAMs, providing a docking site for ZAP 70

(4) ZAP 70 docks on the ITAMs and get phosphorylated by p56lck and activated

(5) ZAP-70 goes on to phosphorylate a variety of molecules, setting off other downstream pathways through transcriptional activation

—DAG-IP3 formed through hydrolysis of membrane phospholipid by phospholipase C, causes ER to release Ca2+

—Ca2+ activates NFAT (up regulates immune response)

—DAG activates NF-κB (up regulates immune response)

—Activation of Ras, the G protein

—Transcription factors activated by MAP kinases turn on genes that trigger cell division

Regulation

B7 is a membrane bound protein on APCs binding either to the CD28 or CTLA-4 of a T cell

CD28 Binding (Upregulation)

The cell will begin proliferating and dividing, transcribing the message for both IL-2 and the high affinity version of its TCR α chain and enter into a positive feedback loop.

*An APC without the B7 for CD28 would turn the T cell off

CTLA-4 Binding (Downregulation)

The T cell will being to downregulate its development

*Resting T cells initially have no CTLA-4, but begin to produce it during activation

Core Concept 8.16: Lipid Rafts

Over the course of the signaling lipid rafts migrate towards the outside of the synapse region as a result of CAMs associating CD45 to CD22

(1) TCR, lipid raft after antigen binding

<FIGURE>

(2) Synapse during mid-activation

<FIGURE>

(3) Synapse, mature (bull's eye formation)

<FIGURE>

9 Cytokines and Signals

9.1 Receptor Signaling Types

Core Concept 9.1: Immunoglobulin

<FIGURE>

Structure Receptors have β pleated sheet (Ig) domains and membrane span characteristics of family

Ligands—antigens, integrins, growth factors

Core Concept 9.2: Type I Receptors

<FIGURE>

Structure

Heterodimers or trimers with interchangeable β and γ subunits across different receptors

Composed of fibronectin domain (like Ig domains), two β pleated sheets running opposite

β,γ N terminals—Four conserved cysteines that stabilize the structure using 2 disulfide bonds

β,γ proximal domains—conserved [tryptophan]-[serine]-[whatever]-[tryptophan]-[serine] sequence (function unknown)

β,γ C terminals—domains have sites that recruit cytosolic tyrosine kinases (phosphorylation)

Ligands—a small monomer with 4 α helical domains

JAK-STAT Pathway—

Ligand binding brings JAKs together and activates

P-JAKs phosphorylate the cytoplasmic receptor tails (docking site for STAT)

JAKs phosphorylate the STATs, which migrate to the nucleus, and up-regulate other genes

Core Concept 9.3: Type II Receptors

<FIGURE>

Structure

β, γ heterodimers with extensive cytoplasmic signaling domains

Fibronectin domain

β,γ N terminals—CCCC that stabilize the structure using 2 disulfide bonds

Ligands—series of α helices

JAK-STAT Pathway—SAME as Type I

Ligand binding brings JAKs together and activates

P-JAKs phosphorylate the cytoplasmic receptor tails (docking site for STAT)

JAKs phosphorylate the STATs, which migrate to the nucleus, and up-regulate other genes

***JAKs and STATs have variety (4 and 6 known respectively) and the signaling process uses more than one version of each

Core Concept 9.4: Interleukin 17

<FIGURE>

Structure

Heterodimers, Homodimers, Trimers

IL17R composed on 2 IL17RAs and 1 IL17RC

Ligands—IL-17A: β pleated sheets held together by 2 S=S bonds constructed by CCCC (pro-inflammatory)

Activates the Act1 Pathway (NF-κB, MAPK, CCAAT)

**IL-17 is secreted by Th17, iNKT, and γδ T cells (mainly for border patrol)

Core Concept 9.5: Tumor Necrosis Factor

<FIGURE>

Structure

Four repeated β pleated sheet domains stabilized by CCCCs' S=S bonds

Upon binding TNF, subunits cluster in trimer, leading to activation

Ligands—juxtacrine factors composed of β pleated sheets and displayed in clusters of three subunits

Pathway binds to Death Domain (DD), TRADD, FADD, and RIP → canonical NF-κB pathway

Core Concept 9.6: Chemokines

<FIGURE>

Structure

7-spanning α helices in membrane associated with G-protein on C-terminal

Very promiscuous receptor (cross reception)

Ligands—α helices and β pleated sheets stabilized with CCCCs

G Protein Pathway—

Ligand binding to receptor induces α subunit of G protein to switch out its GDP to GTP

α, β, γ G Protein subunits all dissociate and start their individual signaling cascades

9.2 Immune Signaling Types & Themes

Core Concept 9.7: Immune Signaling Types

<TABLE>

Core Concept 9.8: Signaling Themes

<FIGURE>

(a) **Phosphorylation**

Adding phosphates to, and removing phosphates from molecules

Examples: BCRs, JAK-STAT, TCRs, G Proteins, Membrane inositols

(b) **Phospholipid Hydrolysis**

Phospholipids are split

Examples:

Phospholipid is split into

DAG (non polar part)

IP3 (polar part) cascades

*IP3 makes the ER leaky to Ca2+, leading to the NFAT pathway through binding to calmodulin

(c) **Protein Hydrolysis**

Using hydrolysis of cytosolic components in situations where you don't want to reverse the signal (once you've clipped a protein, you can't reassemble it, you must synthesize a new one)

Examples: Notch Signaling, Apoptosis Induction

(d) **Cytoskeletal Changes**

In relation to the movement of a cell throughout the body

Examples: Chemokine → 7-spanR → G Protein signal → Integrin conformational change → Talins → Actin microfilaments → Movement

9.3 Cytokine Interaction Types

Core Concept 9.9: Cytokine Interaction Types

(a) **Pleiotropy**

<FIGURE>

One cytokine may act on several different cell types

IL-4 coordinates a Th2 response by acting on both B and mast cells

(b) **Redundancy**

<FIGURE>

More than one cytokine will produce the same effect

IL-2, 4, and 5 induces proliferation in B cells

(c) **Synergy**

<FIGURE>

Combinations of stimulus have effects neither has by itself

IL-4 and IL-5 together will cause B cells to class switch and make IgE immunoglobulins

(d) **Antagonism**

<FIGURE>

One cytokine block the effects of another

IFNγ (a Th1 promoter) blocks the ability of IL-4 (a Th2 promoter) to induce class switching to IgE

(e) **Cascade Inductionk**

<FIGURE>

One cytokine stimulates other cells to produce a different cytokine, which stimulates more cells

Macrophage → IL-12 → Th1 → IFNγ → activated macrophage pathway of development